

Event Registration System Project Book

Public Registration, Admin Review Workflow, Request States, And Audit-Ready Operations

PlanningSoftwares

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Chapter 1: Introduction And System Overview

1.1 Project Title

The project title is **Event Registration System**. The system is designed for a company that organizes an event and needs a controlled way to receive, review, categorize, and manage visitor registration requests.

1.2 Problem Description

Event registration is often handled through simple forms or manual lists. This approach creates several problems:

- Duplicate registrations may be submitted using the same phone number or email.
- Visitor data may be incomplete or invalid.
- Admins may not have a clear workflow for reviewing requests.
- Requests may be difficult to filter by major, city, state, or date.
- There may be no audit record showing who changed a request and when.
- Statistics and exports may require manual work.

The main problem is not only collecting registrations. The real problem is managing registration requests in a professional review workflow where every request has a clear state, valid contact information, and traceable administrative actions.

1.3 Project Objectives

The system aims to:

- Display event registration capabilities professionally.
- Receive registration requests from visitors.
- Validate visitor data before saving it.
- Prevent duplicate registrations by phone number and email.
- Store every accepted request under an initial review state.
- Allow admins to search, filter, sort, and review requests.
- Allow admins to change request states and add internal notes.
- Provide statistics that summarize the registration process.
- Export request data when needed.
- Record important actions in an audit log.

1.4 System Overview

The Event Registration System consists of a public registration workflow and an administrative review workflow.

Visitors submit a form containing their full name, age, major, phone number, email, and city. The backend validates this data, normalizes phone and email values, checks for duplicates, and saves valid requests under the `under_review` state.

Admins log in to a protected control panel. They can view the request table, filter by major, state, city, or registration date, search by name, phone, or email, review request details, change the request state, add notes, archive records, view statistics, and export data.

1.5 Target Users

User	Description	Main Actions
Visitor / Applicant	A person interested in registering for the event.	Submit registration information and receive a request number.
Admin Reviewer	A staff member responsible for reviewing requests.	Log in, review requests, change states, add notes, export data, and view statistics.

Chapter 2: System Analysis And Design

2.1 Functional Requirements

Functional requirements describe what the system must do.

ID	Requirement
FR-01	The system shall allow visitors to submit a registration request.
FR-02	The system shall validate full name, age, major, phone number, email, and city.
FR-03	The system shall normalize email and phone data before duplicate checks.
FR-04	The system shall reject duplicate registrations by normalized phone or normalized email.
FR-05	The system shall create a unique request number for every accepted registration.
FR-06	The system shall save new requests under the <code>under_review</code> state.
FR-07	The system shall allow admins to log in securely.
FR-08	The system shall allow admins to list, search, filter, sort, and paginate requests.
FR-09	The system shall allow admins to view request details.
FR-10	The system shall allow admins to change request state.
FR-11	The system shall allow admins to add internal notes.
FR-12	The system shall allow admins to archive and unarchive requests.
FR-13	The system shall allow admins to manage request states.
FR-14	The system shall prevent deleting linked states unless records are transferred.
FR-15	The system shall provide statistics by state, major, and city.
FR-16	The system shall export filtered request data as CSV.
FR-17	The system shall record important actions in an audit log.

2.2 Non-Functional Requirements

Category	Requirement
Security	Admin endpoints must be protected by authentication. Passwords must be hashed.
Reliability	Accepted registrations must be stored consistently in PostgreSQL.
Maintainability	The backend should be organized using MVC-style separation.
Validation	Invalid inputs must return clear structured errors.
Auditability	Important actions must be stored in action logs.
Performance	Request lists must support pagination and indexed filters.
Extensibility	Request states must be configurable without changing code.
Data Quality	Phone and email values must be normalized for duplicate prevention.

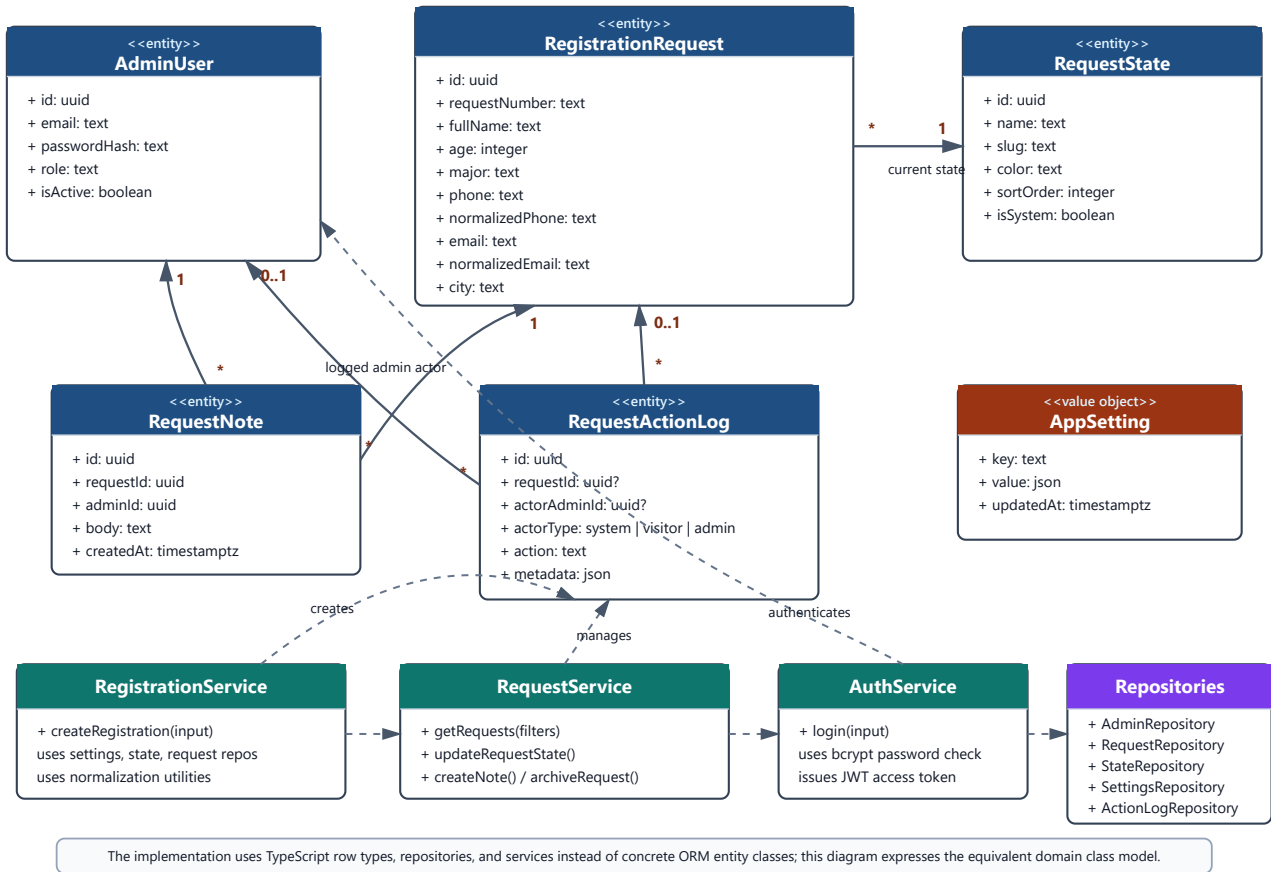
2.3 Use Case Scenarios

Actor	Use Case	Scenario
Visitor	Submit registration request	Visitor opens the registration page, enters valid data, submits the form, and receives a request number.
Visitor	Submit invalid data	Visitor submits missing or invalid values and receives field-level validation errors.
Visitor	Submit duplicate data	Visitor uses an existing phone or email and receives a duplicate registration error.
Admin Reviewer	Review requests	Admin logs in, opens the request table, filters records, and opens a request detail page.
Admin Reviewer	Change request state	Admin reviews a request and moves it from <code>under_review</code> to another state.
Admin Reviewer	Add internal note	Admin records private review comments for future reference.
Admin Reviewer	Export data	Admin exports filtered requests as a CSV file.
Admin Reviewer	Manage states	Admin adds, edits, recolors, reorders, or deletes states according to system rules.

2.4 Class Diagram

The class diagram summarizes the main domain classes concluded from the project implementation and database schema. The backend uses TypeScript row types, repositories, and services rather than ORM entity classes, so the diagram represents the domain model and its service boundaries.

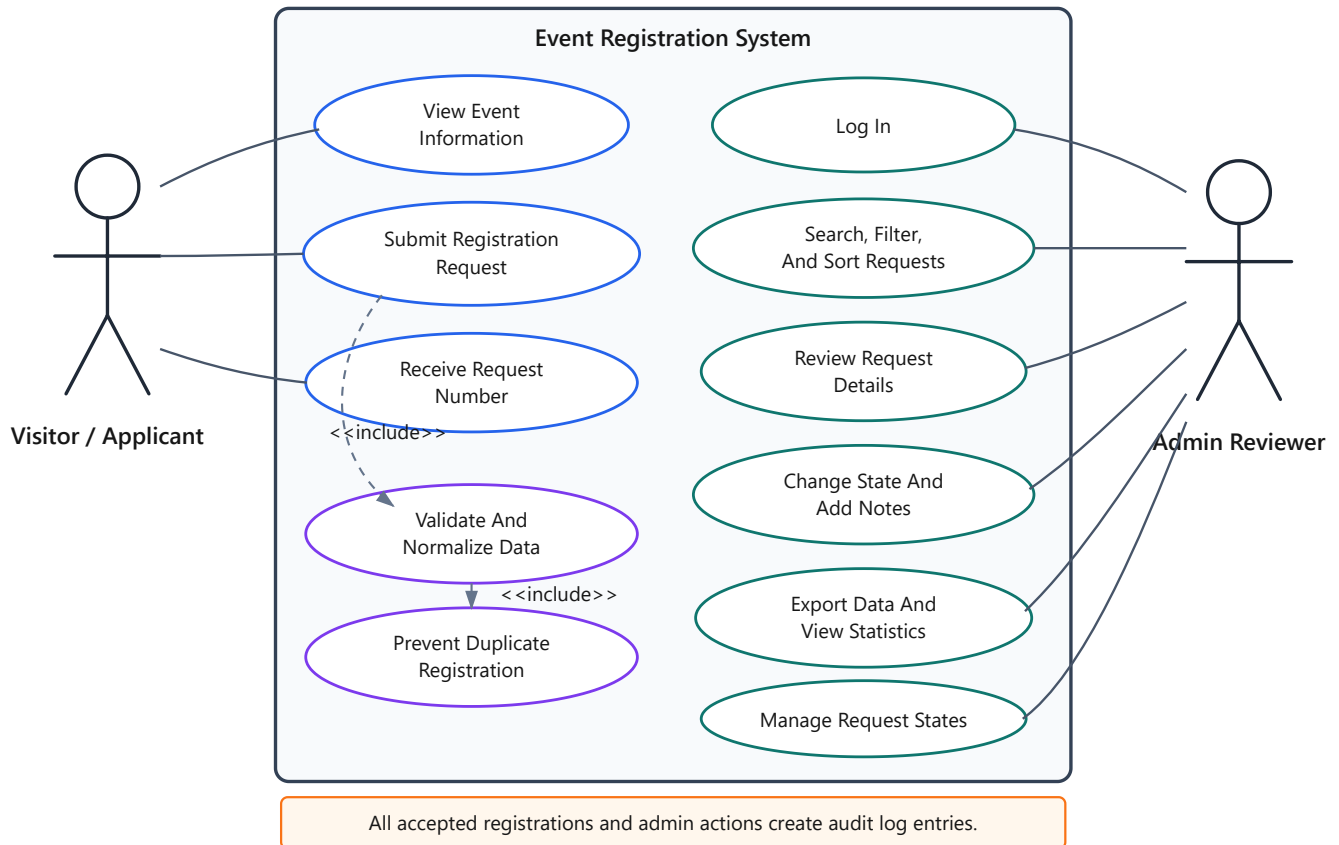
Event Registration System - Class Diagram



2.5 Use Case Diagram

The use case diagram shows the main interactions between the visitor, admin reviewer, and the system.

Event Registration System - Use Case Diagram



2.6 System Architecture

The backend is designed as a Node.js application using Express. PostgreSQL stores the data. The frontend can be a public registration page and an admin control panel that both communicate with the backend API.

Visitor Browser

- > Public Registration Page
- > Express API
- > PostgreSQL Database

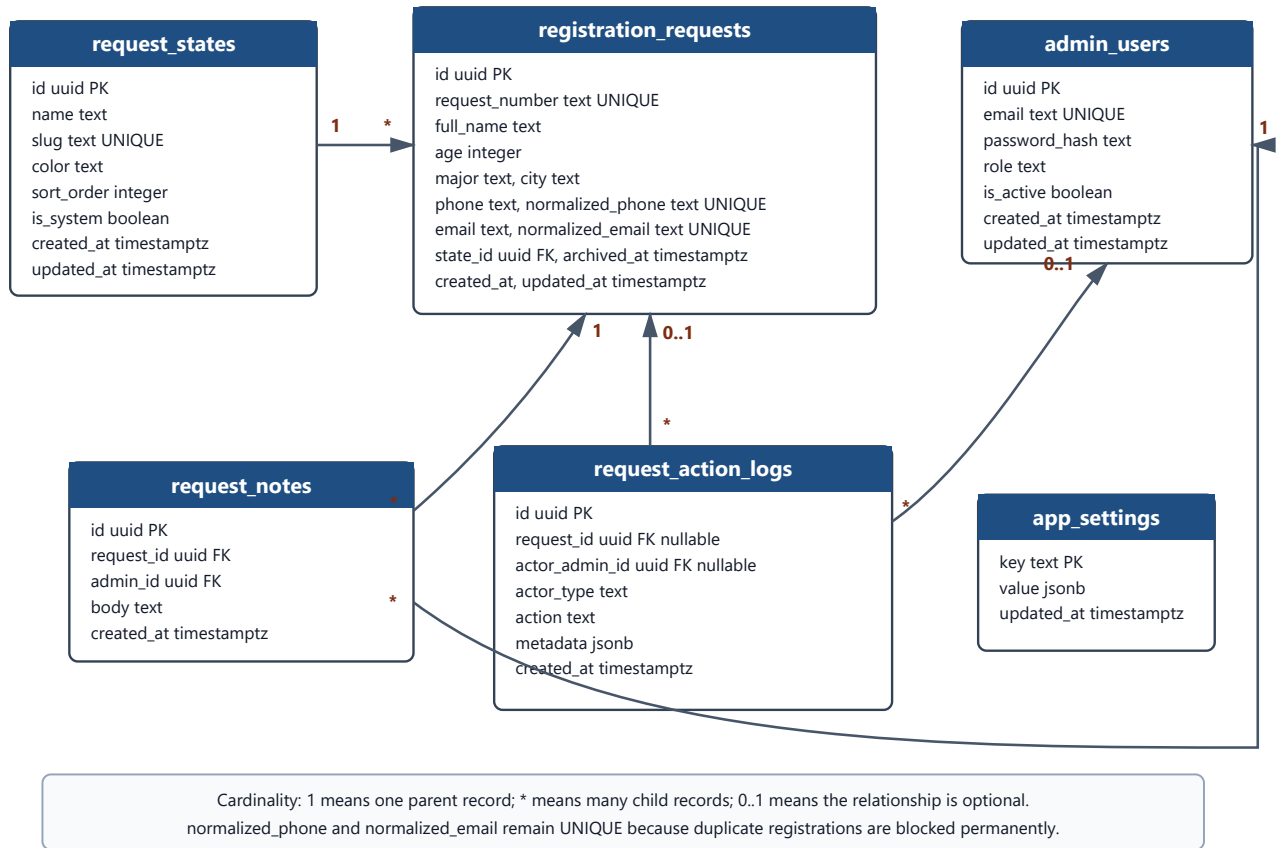
Admin Browser

- > Admin Control Panel
- > Protected Express API
- > PostgreSQL Database

2.7 Database Design

The database is designed around registration requests, request states, admin users, notes, action logs, and settings.

Event Registration System - Database Schema



2.8 Database Tables

Table	Purpose
admin_users	Stores admin accounts, password hashes, roles, and status.
request_states	Stores configurable review states such as under review, approved, and no reply.
registration_requests	Stores visitor registration data and current state.
request_notes	Stores internal admin notes attached to requests.
request_action_logs	Stores audit records for system and admin actions.
app_settings	Stores configurable values such as age limits and default state.

2.9 Table Schemas

2.9.1 admin_users

Column	Type	Description
id	uuid	Primary key.
email	text	Unique admin email.
password_hash	text	Hashed password, not the plain password.
role	text	Initial value is <code>admin</code> .
is_active	boolean	Controls whether the account can log in.
created_at	timestampz	Creation time.
updated_at	timestampz	Last update time.

2.9.2 request_states

Column	Type	Description
id	uuid	Primary key.
name	text	Display name.
slug	text	Unique stable state identifier.
color	text	Color used by the admin interface.
sort_order	integer	Display order.
is_system	boolean	Marks default states that cannot be deleted.
created_at	timestampz	Creation time.
updated_at	timestampz	Last update time.

2.9.3 registration_requests

Column	Type	Description
id	uuid	Primary key.
request_number	text	Human-readable unique request number.
full_name	text	Applicant full name.
age	integer	Applicant age.
major	text	Applicant major.
phone	text	Original submitted phone value.
normalized_phone	text	E.164 phone value used for duplicate checks.
email	text	Submitted email.
normalized_email	text	Lowercase email used for duplicate checks.
city	text	Applicant city.
state_id	uuid	Foreign key to request_states .
archived_at	timestamptz	Archive time, nullable.
created_at	timestamptz	Registration time.
updated_at	timestamptz	Last update time.

2.9.4 request_notes

Column	Type	Description
id	uuid	Primary key.
request_id	uuid	Foreign key to registration_requests .
admin_id	uuid	Foreign key to admin_users .
body	text	Internal note text.
created_at	timestamptz	Note creation time.

2.9.5 request_action_logs

Column	Type	Description
id	uuid	Primary key.
request_id	uuid	Related request, nullable for global actions.
actor_admin_id	uuid	Admin actor, nullable for system or visitor actions.
actor_type	text	system, visitor, or admin.
action	text	Action name such as registration_created.
metadata	jsonb	Extra action details.
created_at	timestampz	Action time.

2.9.6 app_settings

Column	Type	Description
key	text	Primary key for the setting.
value	jsonb	Setting value.
updated_at	timestampz	Last update time.

2.10 Relationships

The schema mainly uses one-to-many relationships. The diagram shows these relationships using 1, *, and 0..1 labels.

Relationship	Type	Explanation
request_states to registration_requests	One-to-many	One state can be assigned to many registration requests. Each request has one current state.
registration_requests to request_notes	One-to-many	One request can have many internal notes. Each note belongs to one request.
admin_users to request_notes	One-to-many	One admin can write many notes. Each note is written by one admin.
registration_requests to request_action_logs	Optional one-to-many	One request can have many action logs. The request link is nullable so some system actions can be logged globally.
admin_users to request_action_logs	Optional one-to-many	One admin can perform many logged actions. The admin link is nullable for visitor or system actions.
app_settings	No direct relationship	Settings are standalone key-value records used by the application.

The `normalized_phone` and `normalized_email` columns remain unique because this project blocks duplicate registrations permanently, even if an older request is archived.

Chapter 3: Implementation Using Node.js With Express

3.1 Implementation Overview

The backend is planned as a Node.js application using Express. Express is responsible for routing HTTP requests to controllers. PostgreSQL is used as the database. The `pg` package is used for database connections and queries. Migrations are managed using `node-pg-migrate`.

The project follows an MVC-style structure:

- **Model:** database tables, validation schemas, repositories, and domain services.
- **Controller:** Express handlers that receive requests, call services, and return responses.
- **View:** API responses consumed by a public registration page and admin control panel. Since this project phase is backend-focused, the view layer is represented by JSON responses and CSV output.

3.2 Suggested Project Structure

```
src/  
  app.ts  
  server.ts  
  config/  
  db/  
  routes/  
  controllers/  
  services/  
  repositories/  
  validators/  
  middleware/  
  utils/  
  migrations/  
  tests/  
  docs/
```

3.3 Model Layer

The model layer represents the data and business rules. In this system, the model is not only one file. It includes:

- PostgreSQL tables.
- Repository functions that query those tables.

- Validation schemas for incoming data.
- Service rules such as duplicate prevention and state deletion protection.

Example model responsibility:

```
type RegistrationRequest = {  
  id: string;  
  requestNumber: string;  
  fullName: string;  
  age: number;  
  major: string;  
  phone: string;  
  normalizedPhone: string;  
  email: string;  
  normalizedEmail: string;  
  city: string;  
  stateId: string;  
};
```

This type helps the developer understand which fields belong to a registration request and how the application should treat the record.

3.4 Controller Layer

Controllers handle HTTP input and output. They should not contain complex business logic. For example, a registration controller receives the request body, passes it to a registration service, and returns a success or error response.

```
async function createRegistration(req, res, next) {  
  try {  
    const result = await registrationService.create(req.body);  
    res.status(201).json(result);  
  } catch (error) {  
    next(error);  
  }  
}
```

The controller is simple because validation, normalization, duplicate checking, and saving are handled by the service and repository layers.

3.5 View Layer

In a traditional MVC system, the view is the user interface. In this backend-focused project, the Express API returns JSON responses that the frontend can display.

Example success response:

```
{
  "requestNumber": "REQ-2026-0001",
  "state": "under_review",
  "message": "Registration request submitted successfully."
}
```

Example validation response:

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Registration data is invalid.",
    "fields": {
      "email": ["Email format is invalid."]
    }
  }
}
```

3.6 Database Connection Using `pg`

The `pg` package is used to connect Node.js with PostgreSQL. A shared connection pool is preferred because it reuses database connections and improves performance.

```
import { Pool } from "pg";

export const pool = new Pool({
  connectionString: process.env.DATABASE_URL
});
```

Repository files use the pool to run parameterized SQL queries. Parameterized queries are important because they reduce SQL injection risk.

```
const result = await pool.query(
  "select * from registration_requests where normalized_email = $1",
  [normalizedEmail]
);
```

3.7 Migrations And Why They Are Used

Migrations are versioned database changes. They are used so every developer and deployment environment can create the same database structure.

This project uses `node-pg-migrate` because:

- It works well with PostgreSQL.
- It keeps database changes organized.
- It supports creating and rolling back tables.
- It makes schema changes repeatable and reviewable.

Example migration idea:

```
export async function up(pgm) {
  pgm.createTable("request_states", {
    id: { type: "uuid", primaryKey: true },
    name: { type: "text", notNull: true },
    slug: { type: "text", notNull: true, unique: true },
    color: { type: "text", notNull: true },
    sort_order: { type: "integer", notNull: true },
    is_system: { type: "boolean", notNull: true, default: false }
  });
}
```

3.8 CRUD Operations

CRUD means Create, Read, Update, and Delete. In this system, CRUD is used carefully because some records should not be physically deleted.

3.8.1 Registration Requests

Operation	Behavior
Create	Visitor submits a valid registration request.
Read	Admin lists and views request details.
Update	Admin changes state or archives the request.
Delete	Normal deletion is not used; requests are archived instead.

3.8.2 Request States

Operation	Behavior
Create	Admin creates a new review state.
Read	Admin lists states for filters and forms.
Update	Admin edits state name, color, or order.
Delete	Admin deletes only unlinked non-system states, or transfers records first.

3.8.3 Internal Notes

Operation	Behavior
Create	Admin adds a note to a request.
Read	Notes appear in request details.
Update	Not required in the first version.
Delete	Not required in the first version because notes are part of the audit context.

3.8.4 App Settings

Operation	Behavior
Create	Default settings are created during migration or seed.
Read	Services read settings such as age limits.
Update	Admin updates configurable policies.
Delete	Not required in the first version.

3.9 Main API Endpoints

Method	Endpoint	Purpose
POST	/api/registrations	Submit public registration request.
POST	/api/auth/login	Admin login.
GET	/api/auth/me	Read current admin profile.
GET	/api/admin/requests	List requests.
GET	/api/admin/requests/:id	View request details.
PATCH	/api/admin/requests/:id/state	Change request state.
POST	/api/admin/requests/:id/notes	Add internal note.
PATCH	/api/admin/requests/:id/archive	Archive or unarchive request.
GET	/api/admin/states	List request states.
POST	/api/admin/states	Create request state.
PATCH	/api/admin/states/:id	Update request state.
DELETE	/api/admin/states/:id	Delete or transfer request state.
GET	/api/admin/stats	Read statistics.
GET	/api/admin/requests/export.csv	Export requests.

3.10 Understanding The Implementation Flow

When a visitor submits a registration:

1. Express receives the `POST /api/registrations` request.
2. The controller sends the body to the service.
3. The service validates the fields.
4. The service normalizes phone and email.
5. The repository checks for duplicate records.
6. If there is no duplicate, the repository inserts the request.
7. The service creates an action log.
8. The controller returns the request number.

This flow keeps the implementation organized and makes it easier to test each part separately.

Chapter 4: Testing And Results

4.1 Testing Method

The system should be tested using a combination of manual API testing and automated tests.

Test Type	Purpose
Unit tests	Test validation, normalization, duplicate checking, and state rules.
API tests	Test Express endpoints using sample requests and responses.
Database tests	Test migrations, constraints, and relationships.
Manual tests	Confirm important workflows from visitor and admin perspective.

4.2 Sample Inputs And Outputs

4.2.1 Valid Registration

Input:

```
{
  "fullName": "Ali Salem Mohamed",
  "age": 24,
  "major": "Architecture Engineering",
  "phone": "+218912345678",
  "email": "ali@example.com",
  "city": "Tripoli"
}
```

Expected output:

```
{
  "requestNumber": "REQ-2026-0001",
  "state": "under_review",
  "message": "Registration request submitted successfully."
}
```

4.2.2 Invalid Registration

Input:

```
{
  "fullName": "Ali",
  "age": 15,
  "major": "",
  "phone": "123",
  "email": "wrong-email",
  "city": ""
}
```

Expected output:

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Registration data is invalid."
  }
}
```

4.2.3 Duplicate Registration

Expected output when the same normalized phone or email already exists:

```
{
  "error": {
    "code": "DUPLICATE_REGISTRATION",
    "message": "A registration already exists for this phone number or email.",
    "conflicts": ["phone", "email"]
  }
}
```

4.2.4 Admin Login

Input:

```
{
  "email": "admin@example.com",
  "password": "secure-password"
}
```

Expected output:

```
{
  "accessToken": "jwt-token",
  "admin": {
    "email": "admin@example.com",
    "role": "admin"
  }
}
```

4.2.5 State Update

Input:

```
{
  "state": "approved"
}
```

Expected output:

```
{
  "requestNumber": "REQ-2026-0001",
  "state": "approved",
  "message": "Request state updated successfully."
}
```

4.2.6 Statistics Response

Expected output:

```
{
  "totalRequests": 120,
  "approved": 45,
  "underReview": 50,
  "noReply": 25,
  "byMajor": [
    { "major": "Architecture Engineering", "count": 20 }
  ],
  "byCity": [
    { "city": "Tripoli", "count": 75 }
  ]
}
```

4.3 Test Cases And Expected Results

Test Case	Expected Result
Submit valid registration	Request is saved and request number is returned.
Submit name with fewer than three words	Validation error is returned.
Submit age below minimum	Validation error is returned.
Submit invalid phone number	Validation error is returned.
Submit duplicate email	Duplicate registration error is returned.
Admin logs in with correct password	JWT access token is returned.
Admin opens request list without token	Unauthorized error is returned.
Admin changes state	State changes and action log is created.
Admin deletes linked state without transfer	Operation is rejected.
Admin exports filtered data	CSV file is returned and export action is logged.

4.4 Problems Faced And Solutions

Problem	Solution
Duplicate registrations can confuse admins.	Normalize phone and email, then block duplicates before saving.
Phone numbers may be written in different formats.	Use a phone-number library to convert valid numbers to E.164 format.
Admins need flexible workflow states.	Store request states in the database instead of hard-coding them.
Linked states cannot be removed safely.	Prevent deletion unless records are transferred to another state.
Sensitive admin routes must be protected.	Use JWT authentication and password hashing.
Database changes must be repeatable.	Use migrations to version schema changes.
Reports need a clean PDF without file path or date footer.	Generate the PDF with browser headers and footers disabled.

4.5 Result Summary

The planned tests prove that the system can receive valid registration requests, reject invalid or duplicate data, protect admin endpoints, manage review states, keep action history, and provide useful statistics and exports.

Chapter 5: Evaluation And Future Improvements

5.1 Project Evaluation

The system design solves the main problem by turning visitor submissions into controlled review requests. It separates public registration from admin operations and gives admins a clear workflow for filtering, reviewing, and updating records.

The design also supports future growth because request states are configurable, settings are stored in the database, and the backend is organized using MVC-style separation.

5.2 Future Improvements

Future versions can add:

- Full public frontend for event details and registration.
- Full admin dashboard interface.
- CMS for event title, description, date, location, and publishing status.
- WhatsApp integration if the organization later decides to send invitations.
- Role-based permissions for multiple admin types.
- Managed tables for majors and cities.
- Email notifications.
- QR code invitation or check-in support.
- Deployment, monitoring, and backup strategy.

5.3 What Was Learned

This project improves understanding of:

- How to analyze a real registration workflow.
- How to design database tables and relationships.
- How MVC separates responsibilities in a backend project.
- How Express controllers, services, and repositories work together.
- Why validation and normalization are important before saving data.
- Why migrations are needed in professional database projects.
- How testing proves that the system works correctly.

Appendix A: Initial Defaults

Item	Default
Minimum age	18
Maximum age	100
Default state	under_review
Duplicate policy	Block normalized phone and normalized email.
Required states	under_review, contacted_awaiting_approval, approved, no_reply.

Appendix B: Out Of Scope For The First Version

- WhatsApp invitation generation or sending.
- CMS or event-content editing.
- Multi-role permission management.
- Managed major and city tables.
- Payment, ticketing, QR codes, or attendance check-in.